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Pushing the Limits: State-Of-The-Art LWD Solutions for Proactive Real-Time Decisions Enabled Drilling the World's ERD Well at 53,000 ft

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Abstract

It is imperative to maximize production while minimizing well time which drove the energy industry to extend the boundaries of Extended Reach Drilling (ERD) aligning it with field development. The steering of longer and more sophisticated wells is crucial for accessing untapped reservoirs and optimizing resource extraction. However, these efforts are accompanied by substantial challenges, such as managing structural uncertainties, ensuring optimal wellbore placement and mitigating downhole torque. The successful drilling of the world's longest ultra-ERD well, which extended to 53,000 feet, showcases how advanced technologies can address these challenges and achieve operational excellence. To overcome these obstacles, a 6-3/4" advanced deep multilayer reservoir mapping solution was deployed as part of an optimized Logging-While-Drilling (LWD) Bottom Hole Assembly (BHA). This system enabled detailed real-time reservoir mapping and enhanced well placement accuracy. The integration of advanced resistivity measurements and inversion mapping facilitated proactive geosteering decisions, ensuring the well remained within the target zone and maintained maximum reservoir contact. A well placement and geosteering decision matrix guided real-time decisions while keeping the wellbore in the optimal zone and avoided geologically complex areas. By maintaining the wellbore in the center of the target zone to avoid dense layers, downhole torque was significantly reduced, improving weight transfer to the bit and enhanced the overall drilling performance. Optimized drilling parameters combined with predictive torque modeling, minimized frictional losses and ensured uninterrupted drilling performance over the extended lateral section. This proactive approach mitigated operational risks and allowed for the precise targeting of high-value zones, ultimately reducing well time and enhancing potential production. The use of LWD BHA optimization, real-time geosteering and advanced reservoir mapping provided a more accurate understanding of the subsurface, enabling better decision-making and reducing drilling inefficiencies. These advancements were instrumental in overcoming the limitations of traditional drilling practices and improved operational efficiency in ultra-ERD operations. This success highlighted the importance of multidisciplinary collaboration, cutting-edge technology and proactive decision-making in addressing the technical challenges of ultra-ERD operations. The integration of advanced reservoir mapping, well placement optimization and torque reduction techniques helped in

setting new benchmarks for future ERD projects. Building on the success of this ultra-ERD well, future initiatives will concentrate on the following aspects which are Enhanced Predictive Models and Torque-Reduction Innovations. These advancements, together with increasing automation and digital technologies, will continue to maximize production, reduce well time and expand the energy industry's capabilities in sustainably developing complex reservoirs.

Introduction

Drilling one of the world's longest horizontal wells has significantly challenged both reservoir insight and drilling solutions. With lateral section up to 37,000 feet long, the interplay between reservoir heterogeneity, mechanical limitations, and operational risk becomes increasingly complex. Success in such ultra-Extended Reach Drilling (ERD) environments hinges not only on the deployment of advanced drilling systems but also on the team's capacity to interpret subtle geological variations, adapt to dynamic subsurface conditions, and optimize performance in real time. By aligning the well trajectory near the reservoir's central axis and steering clear of zones associated with elevated downhole torque, the operation achieved improved weight transfer to the bit and enhanced drilling efficiency. Predictive torque modeling and optimized drilling parameters helped minimize frictional resistance, reducing operational risk, decreasing overall well delivery time, and maximizing reservoir exposure. This case study highlights the critical role of integrating real-time geosteering, deep formation mapping, and advanced LWD technologies in overcoming the technical limitations of conventional ERD practices. These innovations collectively enhance subsurface understanding, support informed decision-making, and mitigate drilling inefficiencies. Future developments in ERD workflows will likely emphasize enhanced predictive modeling, advanced torque-reduction strategies, and greater integration of automation and digitalization—key enablers for the efficient and safe development of complex, high-value reservoirs.

Historical limitations of conventional Geosteering solutions in ERD wells

In this field, conventional Logging-While-Drilling (LWD) technologies have historically posed limitations. These traditional tools are predominantly reactive in nature, offering only shallow depths of investigation. As a result, wells often penetrate non-reservoir or dense intervals before corrective action can be taken. Field experience has shown that, when the trajectory intersects unintended layers, it typically requires a minimum of 500 feet of drilling to steer the wellbore back to the optimal reservoir zone—an outcome that negatively impacts both efficiency and reservoir contact. Additionally, ERD operations in this field must navigate a crowded subsurface environment marked by high-density well spacing and anti-collision constraints ([Figure 1](#)). To prevent reservoir communication and avoid wellbore collapse or fluid loss, operators are frequently required to shift trajectories above or below producing intervals. Compounding this complexity, the target reservoir is relatively thin, with net pay thicknesses ranging from 6 to 10 feet. This demands exceptional trajectory accuracy to maintain wellbore placement near the structural centerline, where even minor dip changes can result in rapid reservoir exit (Nashat et al 2022).

Furthermore, maintaining an optimal mud weight window is critical. The balance must be carefully managed to ensure efficient hole cleaning and pressure support, while avoiding surpassing the fracture gradient—leading to potential formation damage and losses—or falling below formation pressure, which risks wellbore instability and failure.

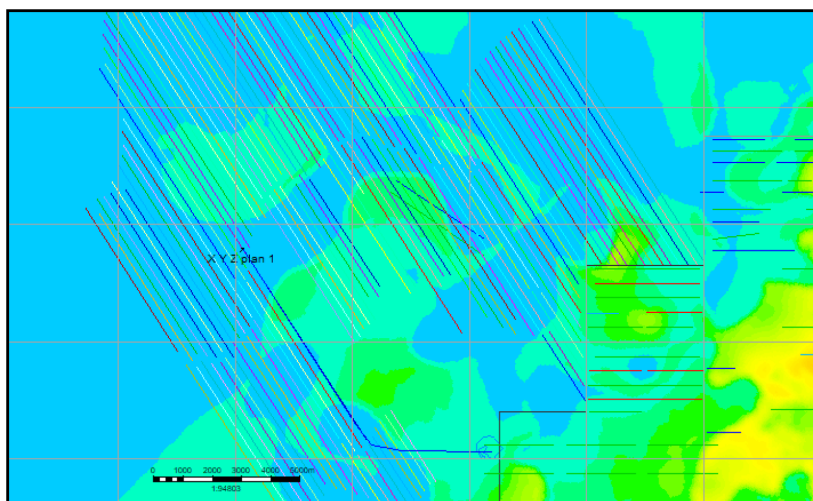


Figure 1—Planned wells with spacing 250 meters. Blue color trajectories refers to injector wells while the red color trajectories refer to oil producer wells

New simplified LWD platform

The new system represents a transition from a modular design to an integrated design, reducing the required number of drill collars from six to two, as shown in Figure 2. The system also features faster processing and communications capabilities than previous designs. These changes bring several benefits, including: (Fares et al 2022)

- A reduction of 50% in the number of connections between collars, dramatically reducing BHA handling time
- A similar reduction in the number of internal components, reducing the number of potential failure points and improving reliability
- A reduction in overall length, bringing sensors closer to the bit and reducing shipping costs
- The ability to program the system in the workshop prior to the job, virtually eliminating time spent communicating with the tool on the drill floor

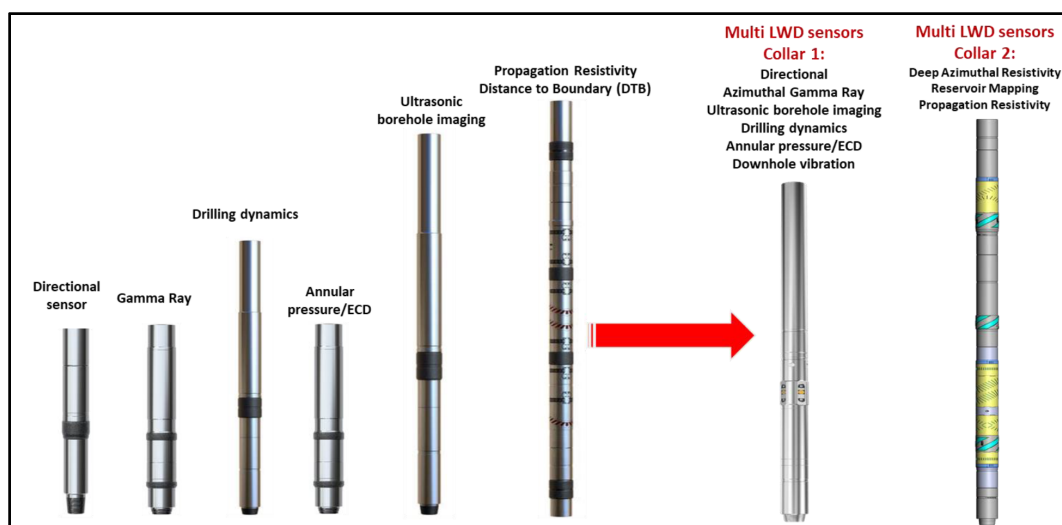


Figure 2—New LWD platform versus Legacy LWD

In this case study, a new-generation Deep Azimuthal Resistivity tool was run with the base collar and Density/Neutron porosity collars.

Deep Azimuthal Resistivity and reservoir mapping solution

To address the ultra-ERD challenges and limitations in conventional Geosteering methodology (reactive way), a new Logging while drilling reservoir mapping solution introduced using the latest generation of Deep Azimuthal Resistivity (DAR) technology [Figure 3](#). This advancement marked a significant step change in the field's geosteering capabilities by enabling proactive, forward-looking reservoir navigation. ([Fares et al 2024](#))

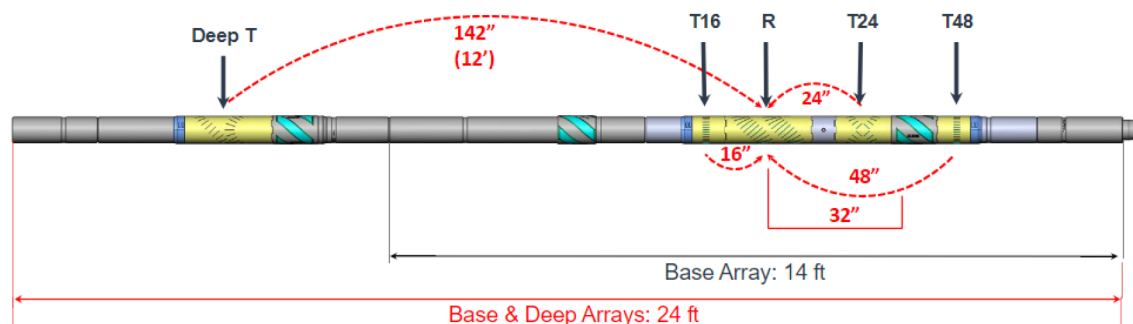


Figure 3—Deep-Azimuthal-Resistivity (DAR) tool design for 1D Inversion capabilities

The main difference lies in the tool physics, in which DAR is equipped with Transmitter and Receiver in the same collar ([Figure 3](#)). Nevertheless, the Transmitter has the latest electrical design to attain the raw data component for 1D-inversion processing. Orthogonal transmitters and receivers come with advanced coil design and low-noise electronics, positioned to provide data from Deep and Base Arrays. Raw components come from Deep Array which are equipped with the capability of transmitting 62.5 kHz, 125 kHz and 250 kHz frequencies. A sophisticated processing algorithm inverts the data and maps the positions, thicknesses, and resistivities of interbedded rock and fluid layers. Real time visualization of the surrounding geology and fluids provides key information required to precisely place the well and maximize reservoir contact. Real-time inversion and imaging capabilities of this tool allowed the geosteering team to dynamically adjust the trajectory in response to evolving subsurface insights. This proved particularly critical in carbonate reservoirs characterized by thin pay, fracture networks, and subtle stratigraphic variations. By maintaining the wellbore in the optimal zone and avoiding premature exits, the team improved reservoir exposure, reduced downhole torque, and enhanced drilling efficiency. The DAR also provides multi-frequency propagation resistivity measurements to deliver a comprehensive understanding of resistivity across the widest range of fluids and rocks. The innovative design of the proprietary crossed transmitter antenna enables the computation of anisotropy in real time, enhancing formation characterization. Real-time access to R_v and R_h in a relatively undisturbed environment drives a more accurate calculation of the water saturation, further supporting advanced petrophysical analysis of the reservoir. This paper presents a successful execution of this ultra-Extended Reach Drilling (ERD) well in well A in Giant offshore carbonate field, implemented an integrated, multidisciplinary workflow supported by prewell modeling and real-time operational alignment across all technical domains.

Well Planning

The well planning workflow was designed to align geological objectives with mechanical constraints while incorporating risk mitigation strategies essential for the success of an ultra-Extended Reach Drilling (Ultra-ERD) operation. A comprehensive and meticulously structured planning phase was initiated well in advance of drilling, laying the groundwork for the successful execution of the 53,000 ft well—currently the longest in the field. This planning effort was part of a long-term strategic roadmap aimed at gradually extending ERD capabilities, which had already pushed measured depths from 45,300 ft to 52,000 ft across previous campaigns. The current Ultra-ERD well represents the culmination of this progressive expansion, with all

key lessons from earlier operations systematically incorporated into both the planning and execution phases. A multi-disciplinary team comprising drilling engineering, survey management, and subsurface disciplines collaborated to design an optimized well trajectory. This involved the integration of offset well data, detailed anti-collision assessments, torque and drag simulations, and casing design validation. A particular focus was placed on minimizing wellbore tortuosity to ensure effective weight transfer and reduce mechanical risks over the extended lateral section.

The subject Well-A was designed as an oil producer well in Reservoir B. [Figure 4](#) The well is located in a tight reservoir area (Avg Permeability 2 mD) which added the challenge to overcome the required injection rate. The well plan is presented in [Figure 4](#), As per well objectives, the BHA design was optimized to achieve effective well-placement operations by placing LWD sensors closer to bit, whilst ensuring a balanced BHA stabilization to ensure meeting directional objectives and minimize vibration. (Nashat et al 2022)

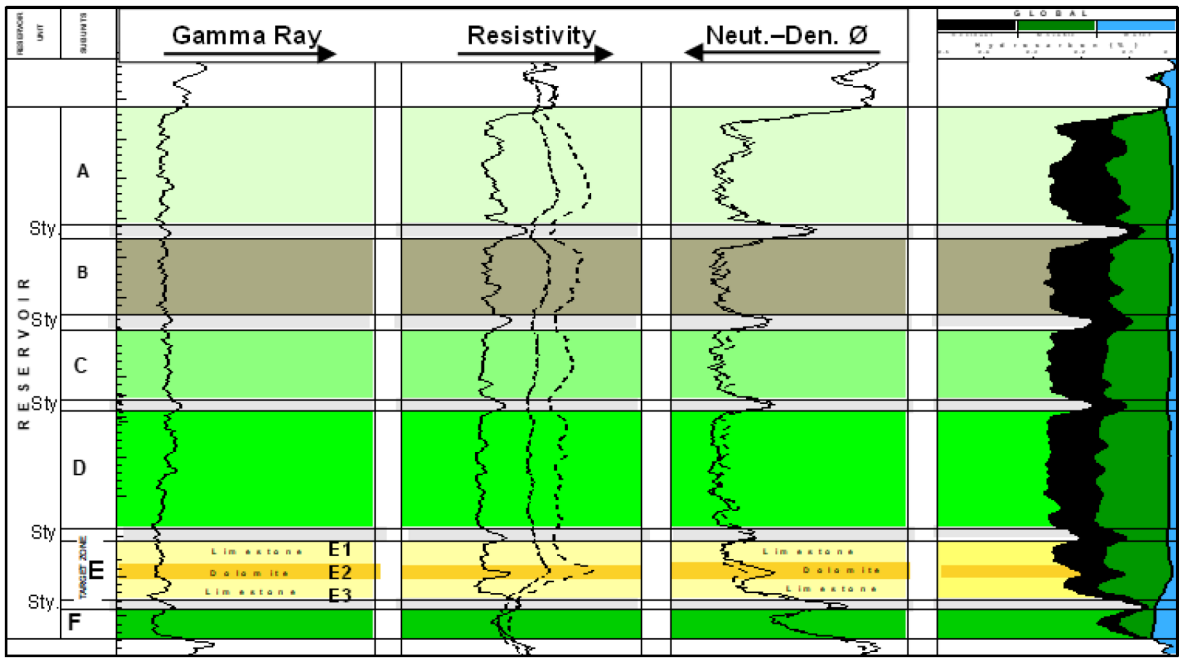


Figure 4—Reservoir layers A, B, C, D, E and F are separated by thin, low-porosity stylolites

After designing the BHA, various sets of models were designed comparing different scenarios to provide the most safe and effective window of drilling parameters.

The models used torque and drag trends, hydraulics, and with the advantage of having the downhole data, dynamic buckling models were also used to ensure efficient transfer of the energy to the bit without exposing the string to undesirable stresses. [Figure 5](#)

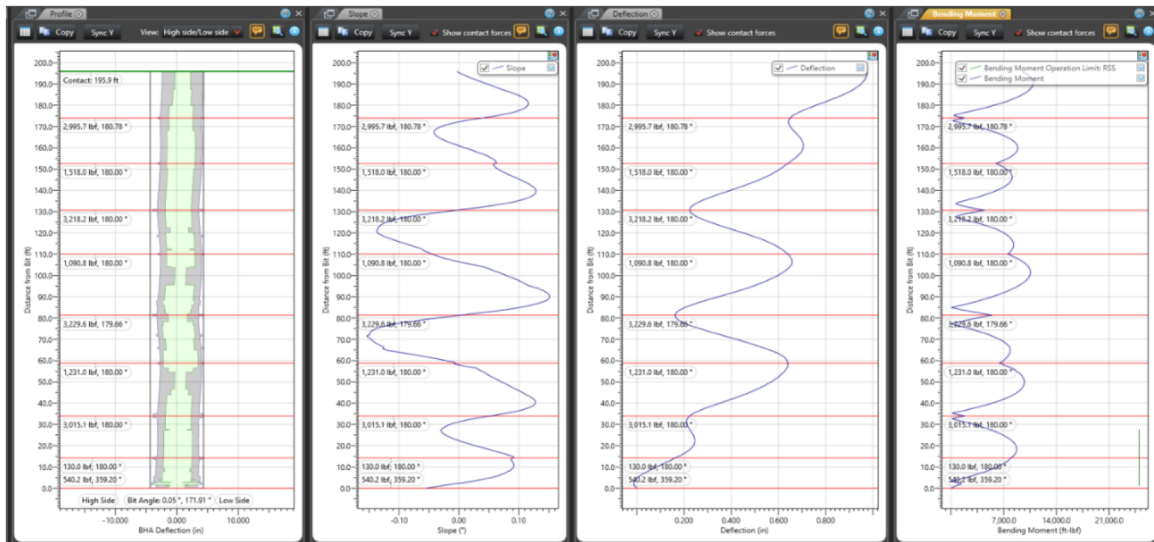


Figure 5—BHA analysis and simulation

The planned trajectory was catered to achieve both drilling and Geosteering objectives and considered avoidance of collision with nearby wells (Figure 6). An important factor for success was comprehensive planning and close collaboration between all parties, considering the challenges associated with high MD/TVD ratios, and defining the appropriate execution road map with risk analysis and mitigation methodologies.

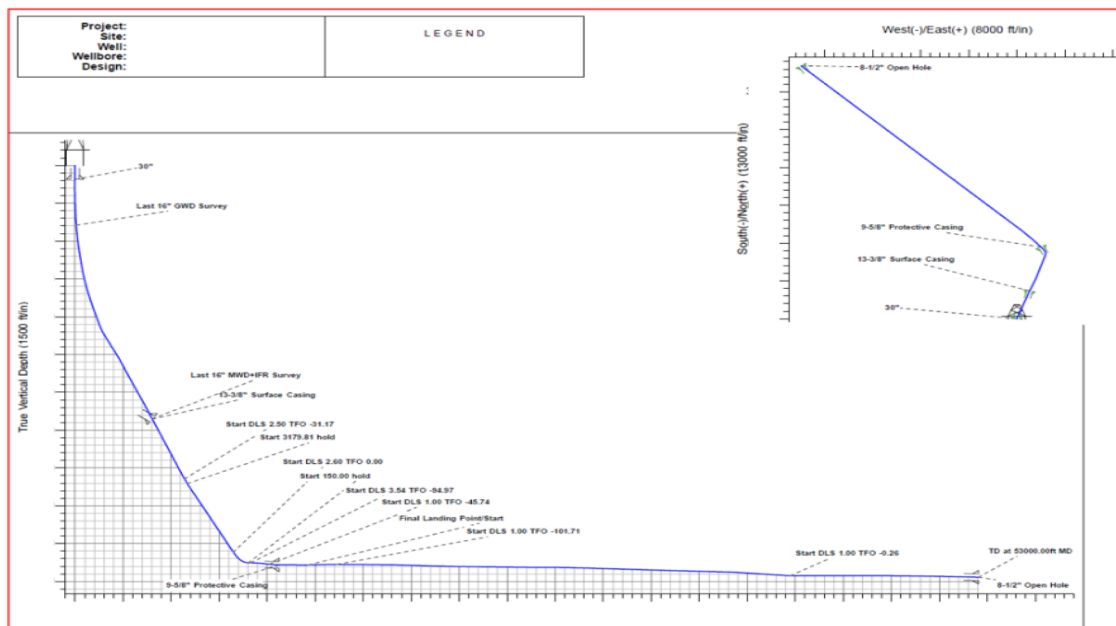


Figure 6—Well-A Planned trajectory

The BHA included Multi LWD collar 1 to provide the following measurements (Directional - Gamma Ray Ultrasonic caliper - Drilling dynamics - Annular pressure/ECD -

Downhole vibration), Multi LWD sensors collar 2, to provide Deep reservoir mapping and realtime 1D inversion along with multi frequency propagation resistivities. In addition to, density, Neutron porosity and formation pressure tester. Intelligent Push the bit rotary steerable system was part of the BHA (Figure 7).

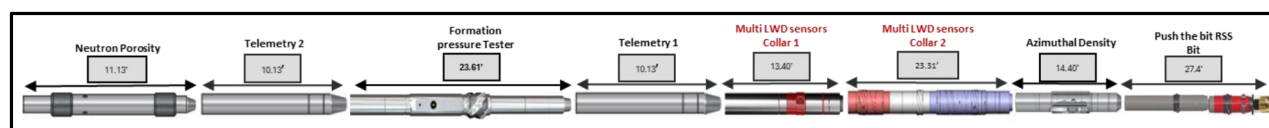


Figure 7—8 1/2" Lateral section BHA configuration

In addition, advanced geological and geophysical modeling was conducted to define structural uncertainty envelopes and reservoir target windows. High-resolution prewell modeling using 1D inversion responses were conducted and have considered all geological scenarios, enhancing pre-drill confidence in DAR deployment and geosteering strategy (Figure 8).

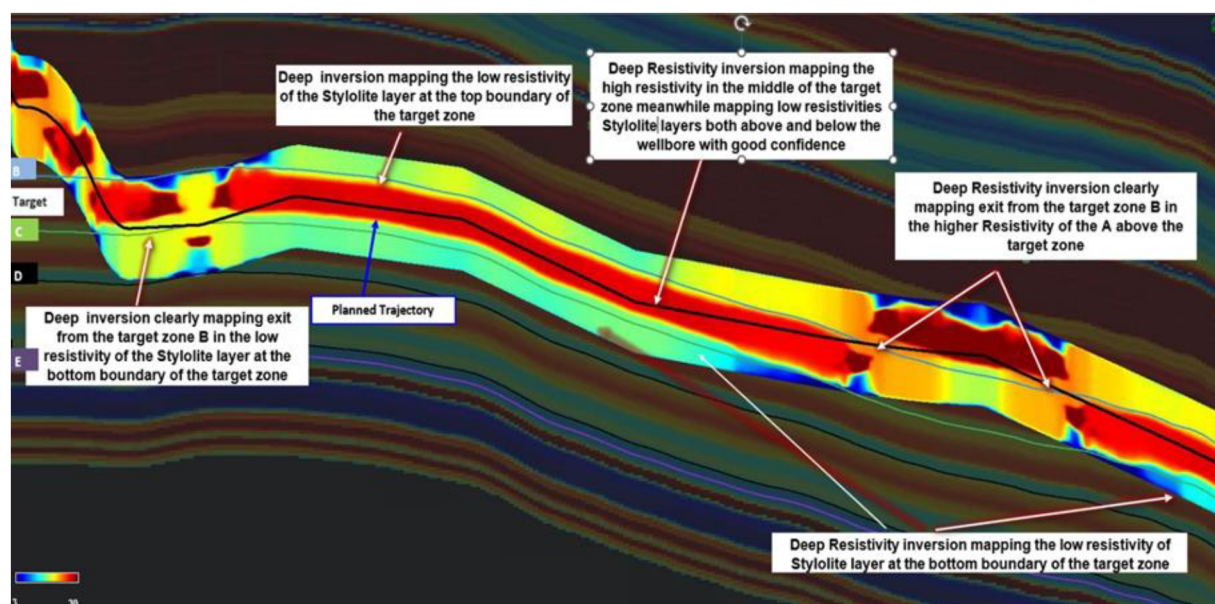


Figure 8—Deep Azimuthal Resistivity Inversion prewell modelling for 8.5in section. 20ft DOI

Drilling Phase

The 8 1/2" lateral section started drilling and mapping successfully both reservoir boundaries. The formation dipping was estimated using the DAR reservoir mapping and was validated using the density image to confirm the optimum Geosteering inclination. The well trajectory was precisely maintained away from the poor porosity boundaries which helped in reduce the downhole torque. Geosteering decisions were primarily driven by subtle inversion responses from the DAR tool, which provided boundary detection up to 20 feet above and below the well trajectory. These insights were continually validated using variations in gamma ray, resistivity, density-neutron, and density image logs. Figure 9. The structural behavior of the reservoir along the lateral section was highly variable and often unpredictable. Dip angles fluctuated throughout the well path, necessitating continuous refinement of the structural model, which was initially based on seismic interpretation and offset well data.

The formation initially appeared flat to gently undulating. A 0.1° up-dip trend over the first 500 feet was followed by a flat segment, which then transitioned into a 0.15° down-dip and subsequently shifted to a 0.2° up-dip. Although minor, these undulations posed a risk of exiting the reservoir if not carefully managed.

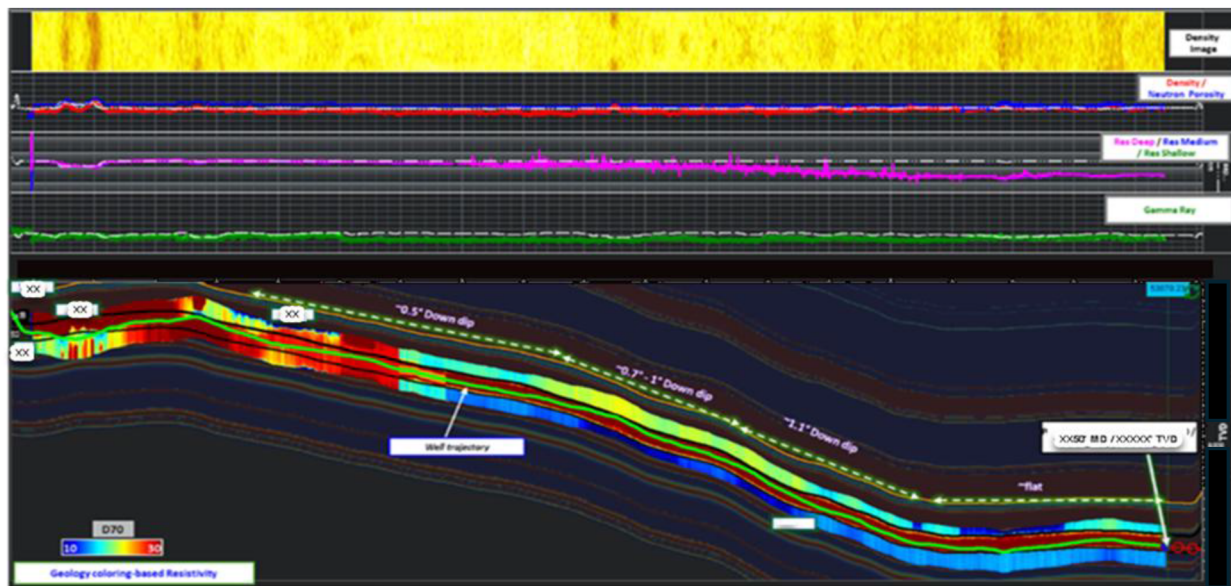


Figure 9—8 1/2" Geosteering cross section using DAR reservoir mapping

As the well steer a regional down-dip trend of 0.4° to 0.6° was observed, complicated by localized structural irregularities likely associated with subtle faulting or intra-reservoir deformation. The dip flattened slightly to 0.2° – 0.3° , providing temporary stability for geosteering operations. In the mid way, the dip steepened significantly to 0.7° – 1.1° , necessitating a reduction in inclination and careful adjustment of tool face orientation to maintain the trajectory within the reservoir, dip started to increase from 1.1° – 1.3° down-dip trend, increasing the risk of drilling into dense formations above the reservoir top. Additional inclination reduction was required to remain within the reservoir's sweet spot. The last 4000 ft structural dip decreased again to approximately 0.1° , but the narrow vertical window of the reservoir continued to pose a geosteering challenge.

These structural complexities demanded precise interpretation and real-time steering decisions to ensure optimal wellbore placement within the target zone throughout the entire lateral section.

Structural interpretation and wellbore placement were validated against the model. Updates were made to the geological framework and tool response models for future wells.

This structured, collaborative approach ensured that subsurface uncertainties were mitigated through robust planning, real-time responsiveness, and comprehensive post-job analysis. By uniting disciplines through a unified workflow, the project achieved superior reservoir contact, minimized risk, and delivered enhanced operational performance.

Results

Drilled, Geosteered and completed the longest ultra-ERD well in the world reaching total depth of 53,000 feet. The optimized solutions helped in maximize the reservoir contact to 100% keeping the well in middle of target zone, reduce the downhole torque which enabled to maintain on bottom ROP above 100 ft/hr - high signal to noise ratio enabled LWD signal detection for reliable real-time data - smooth wellbore enabled tripping in / out of hole and minimized the ILT. The well achieved a net effective pay of 36,532 feet, with an average effective porosity of 20% and average water saturation of 24% Through optimized drilling and geosteering strategies. The DAR has provided accurate 1D inversion results that aligned closely with those from independent logging tools while offering enhanced geological insight. The DAR delineated multiple resistivity boundaries and revealed structural dip variations and lateral heterogeneities within the formation, delivering a clear and intuitive subsurface image. These capabilities enabled proactive

geosteering decisions, allowing the trajectory to effectively avoid dense, non-reservoir layers and remain within the optimal pay zone.

Furthermore, AI-driven surface signal processing, enabled by automated signal detection solution, significantly enhanced real-time signal detection and interpretation, strengthening operational responsiveness.

High signal-to-noise LWD tools ensured dependable real-time formation evaluation, while the smooth borehole profile contributed to efficient tripping operations and minimized invisible lost time (ILT). The well was successfully delivered 14 days ahead of the Approved for Expenditure (AFE) as showed in Figure 10.

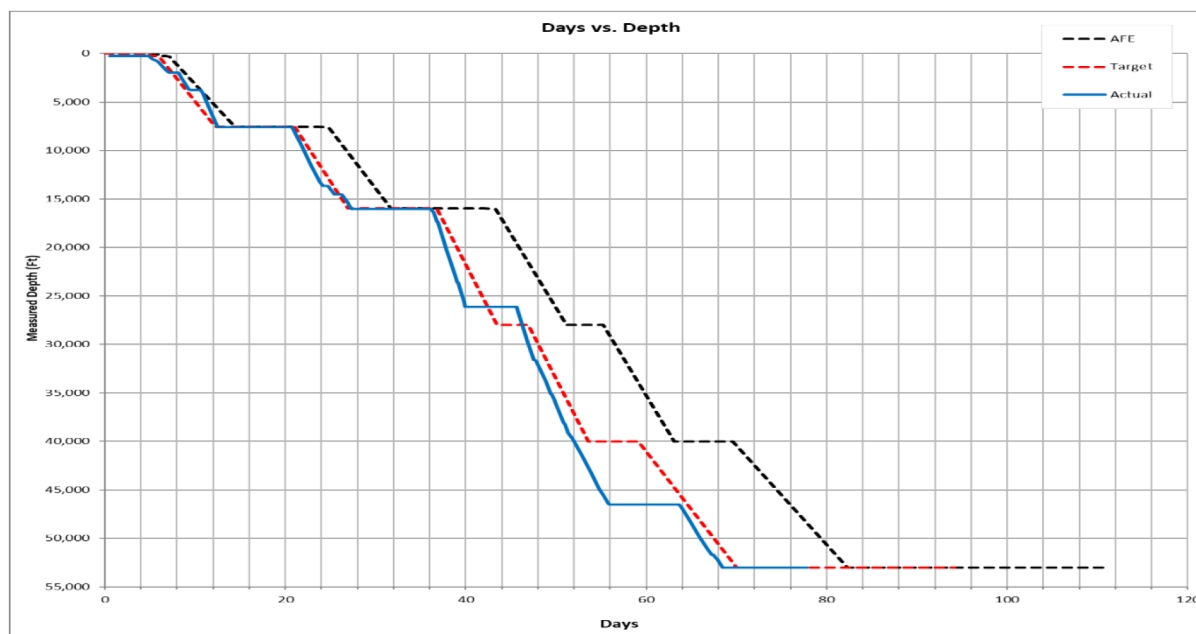


Figure 10—Total footage Daily Progress in MD

Conclusion

The successful delivery of the world's longest ultra-Extended Reach Drilling (ERD) well, reaching a total measured depth of 53,000 feet and achieving a net effective pay of 36,532 feet, exemplifies a major advancement in ERD and real-time geosteering technology. This achievement was made possible through the deployment of latest LWD integrated suite of technologies, most notably the DAR solution for proactive Geosteering decisions. By enabling real-time mapping of formation boundaries up to 20 feet from the borehole and providing detailed inversion imaging, facilitated proactive geosteering and ensured 100% reservoir contact, with the wellbore consistently maintained within the center of the target zone.

The AI-enhanced surface signal processing (Max Receiver), and advanced MWD/LWD telemetry workflows supported robust real-time data transmission and interpretation over a 35,000-ft lateral. These technologies worked synergistically to deliver actionable insights for trajectory control, resistivity contrast mapping, formation anisotropy evaluation, and fluid contact identification. Their integration into a multidisciplinary decision-making framework allowed seamless operational alignment and risk mitigation across all phases of planning and execution.

Furthermore, the deployment of a structured geosteering decision matrix and torque-reduction strategies helped optimize borehole quality, improve ROP, and reduce downhole torque. The well was completed 14 days ahead of the approved AFE and demonstrating the economic value of precision well placement and drilling efficiency in complex carbonate reservoirs.

This case study confirms that a fully integrated, technologically advanced geosteering workflow can overcome conventional LWD limitations, reduce drilling inefficiencies, and set new performance benchmarks in ultra-ERD operations. It underscores the strategic value of combining novel deep resistivity tools, predictive modeling, structured workflows, and cross-functional collaboration to unlock new frontiers in reservoir accessibility, data quality, and economic performance. These findings provide a validated blueprint for future ultra-ERD projects in geologically challenging environments.

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